

KEERTHI CHAKRI

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SUMMARY

Backend Engineer turned AI Engineer with production experience scaling enterprise APIs and secure LLM infrastructure. I spent nearly three years building FastAPI services and Go microservices for **50,000+ active users**, then applied that systems foundation to AI engineering in the healthcare SaaS space. My recent work focuses on making GenAI reliable and compliant—architecting **prompt injection defenses**, validating **HIPAA-aligned security** protocols, and cutting **LLM costs by 50%** while dropping latency to 50ms.

EXPERIENCE

SmileSyncAI

May 2026 - Present

Full Stack Engineer Intern

Paia, HI, USA

- Improved Mobile PageSpeed score from **46** → **91+** by optimizing hero image loading and deferring third-party scripts, maintaining SEO and accessibility scores **above 95/100**.
- Resolved a Next.js CSR bailout by refactoring localStorage-based language context to a **cookie-based SSR-compatible implementation**, eliminating schema markup errors across 147 static pages and reducing validator issues to zero.
- Cut main-thread blocking by **2+ seconds** by replacing external font and script dependencies with local Next.js assets, reducing time to interact across all production routes.

ReplyQuickAI

Jan 2026 - April 2026

Backend Engineer Intern

Sarasota, FL, USA

- Built a production QR routing system using TypeScript and Next.js, mapping **1,000+ concurrent** scans to dynamic dashboards with zero routing errors working in an agile sprint cycle.
- Validated **HMAC security middleware** that cryptographically signs every kiosk-to-backend request, ensuring zero unauthorized access across all HIPAA-regulated patient sessions.
- Improved dental scan quality by tightening image capture rules for all 5 required views — auto-capture now only triggers when teeth are correctly positioned, eliminating off-target submissions reaching the ML model.

Exotel

June 2021 - Nov 2023

Backend Engineer

Chennai, India

- Built REST API backends in Python and FastAPI for 3 enterprise client platforms, handling authentication, data ingestion, and reporting for 50,000+ daily active users with consistent **sub-200ms response times**.
- Rewrote a Python event-routing service in Go using net/http to address throughput limits — the Go version handled over **4x the request volume** with lower memory usage, which resolved recurring timeout issues under peak load.
- Reduced average **API response time by 42%** on high-read reporting endpoints by adding a Redis caching layer with TTL-based invalidation, significantly dropping database load during peak hours.
- Designed PostgreSQL schemas with row-level security and PgBouncer connection pooling for multi-tenant client isolation, establishing a core database architecture blueprint adopted by the department.
- Deployed services on AWS EC2 and RDS with Docker, establishing the **initial CI/CD pipeline infrastructure** utilized across the engineering team.

PROJECTS

LLM Gateway — GenAI Platform Service: Cost Control, Moderation & Evaluation

[Github](#) | Mar 2026 - Present

- Cut API token costs by 50% across 4 OpenAI and Claude models by engineering a pgvector semantic cache that served repeated queries in **50ms at \$0 cost**.
- Simulated and neutralized live prompt injection attacks across 6 tenants using a cosine-similarity intent detection layer that flagged adversarial inputs at a **0.654 threshold**.

Credit Risk ML — Loan Default Prediction System

[Github](#) | April 2026

- Trained and deployed an XGBoost and LightGBM ensemble model on 307,000 credit applications, achieving 84% accuracy and exceeding the target baseline AUC by 9 points.
- Served real-time predictions via a FastAPI endpoint with strict Pydantic validation, maintaining zero failures and a **219ms average latency** under 30-second Locust stress tests.

SKILLS

Languages & Core Frameworks: Python, Go (net/http), TypeScript, SQL, Bash, FastAPI, Next.js, Node.js

Databases & Data Stores: PostgreSQL (RLS, PgBouncer), Redis, Supabase, MongoDB

Backend, DevOps & Infrastructure: REST APIs, AWS (EC2/Lambda/S3/RDS/ECS/CloudWatch), Docker, GitHub Actions CI/CD

AI, LLM & Machine Learning: LangChain, LangGraph, RAG Pipelines, Agentic Workflows, OpenAI API, FAISS, Pinecone, PyTorch, Hugging Face, Scikit-learn, LangFuse

EDUCATION

University of Central Missouri | Master of Science, Computer Science

Jan 2024 - Dec 2025